

# Circuit-Based Quantum Computing

Session 2 – Day 1

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- Study magic states and the magic state injection protocol.

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- Measurements in the computational basis  $\{|x\rangle\}$  extract classical outcomes.

# Elementary Single-Qubit Gates

## Matrix Representations

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad S = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$$

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- **Pauli Gates** ( $X, Y, Z$ ) represent  $\pi$  rotations on the Bloch sphere.
- **Hadamard** ( $H$ ) creates equal superpositions:  $H|0\rangle = |+\rangle$ ,  $H|1\rangle = |-\rangle$ .
- **Phase** ( $S$ ) and  $\pi/8$  **Phase** ( $T$ ) gates:  $T^2 = S$ ,  $S^2 = Z$ .

# Elementary Multi-Qubit Gates

## Matrix Representations

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} = |0\rangle\langle 0| \otimes I + |1\rangle\langle 1| \otimes X$$

$$\text{SWAP} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = |00\rangle\langle 00| + |01\rangle\langle 10| + |10\rangle\langle 01| + |11\rangle\langle 11|$$

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- **CNOT**: Flips target if control is  $|1\rangle$ . Generates entanglement.
- **SWAP**: Exchanges states of two qubits.

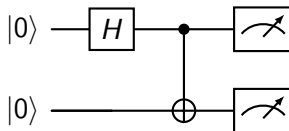
# Drawing Circuits with quantikz

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## Example: Bell State Preparation Circuit



# Bell State Preparation

Preparing the maximally entangled EPR state  $|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ :

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❸ **CNOT (Qubit 0 to Qubit 1):**

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❹ **Measurement:** Yields outcomes 00 or 11 with 50% probability each.

# Superdense Coding – Overview

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Transmit two classical bits ( $ab \in \{00, 01, 10, 11\}$ ) from Alice to Bob by sending only a single physical qubit, using a pre-shared Bell state.

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## Setup:

- Alice and Bob share:  $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle_{AB} + |11\rangle_{AB})$ .
- Alice performs a local operation on her qubit A based on her classical bits.
- Alice sends her single qubit A to Bob.

# Superdense Coding – Encoding & Decoding

## Alice's Encoding:

| Bits | Alice's Gate | Resulting Joint State                                                |
|------|--------------|----------------------------------------------------------------------|
| 00   | $I$          | $ \Phi^+\rangle = \frac{1}{\sqrt{2}}( 00\rangle +  11\rangle)$       |
| 01   | $X$          | $ \Psi^+\rangle = \frac{1}{\sqrt{2}}( 10\rangle +  01\rangle)$       |
| 10   | $Z$          | $ \Phi^-\rangle = \frac{1}{\sqrt{2}}( 00\rangle -  11\rangle)$       |
| 11   | $XZ$         | $-\lvert\Psi^-\rangle = \frac{1}{\sqrt{2}}( 10\rangle -  01\rangle)$ |

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## Bob's Decoding:

- Bob applies CNOT from A to B, then  $H$  on A.
- Bob measures in computational basis, retrieving  $ab$  with 100% certainty.

## The Problem

Given a black-box function  $f : \{0, 1\} \rightarrow \{0, 1\}$ , determine whether  $f$  is **constant** ( $f(0) = f(1)$ ) or **balanced** ( $f(0) \neq f(1)$ ) in a single query.

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**Oracle definition ( $U_f$ ):**

$$U_f |x\rangle |y\rangle = |x\rangle |y \oplus f(x)\rangle$$

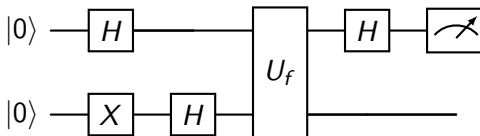
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**Quantum Circuit using Phase Kickback:**



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## 4 Apply $H$ to Qubit 0:

$$H(\pm |-\rangle) = \pm |1\rangle \implies |\psi_3\rangle = \pm |1\rangle |-\rangle$$

Measuring qubit 0 yields **1 with 100% probability.**

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- **Solovay–Kitaev Theorem:** Any single-qubit unitary can be compiled into a sequence of Clifford+ $T$  gates of length:

$$\mathcal{O}(\log^c(1/\epsilon))$$

to achieve an approximation error of  $\epsilon$ , where  $c \approx 3.97$ .

# Clifford vs. Non-Clifford Operations

The Clifford Group ( $\mathcal{C}_n$ ): The normalizer of the Pauli group:

$$\mathcal{C}_n = \{U \in \mathcal{U}(2^n) \mid UPU^\dagger \in \mathcal{P}_n \text{ for all } P \in \mathcal{P}_n\}$$

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**Non-Clifford Gates:** E.g., the  $T$  gate, Toffoli gate. They are necessary to achieve quantum advantage.

- **Eastin–Knill Theorem:** Non-Clifford gates (like  $T$ ) cannot be implemented transversally (fault-tolerantly) in quantum error-correcting codes.

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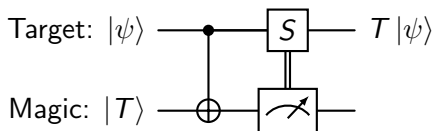
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## Magic State Injection Circuit:



# Magic State Injection – Mathematical Proof

Let  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$  and  $|T\rangle = \frac{1}{\sqrt{2}}(|0\rangle + e^{i\pi/4}|1\rangle)$ .

## ① Initial Joint State:

$$|\Psi_0\rangle = \frac{1}{\sqrt{2}} \left( \alpha|00\rangle + \alpha e^{i\pi/4}|01\rangle + \beta|10\rangle + \beta e^{i\pi/4}|11\rangle \right)$$

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- **Outcome  $m = 1$ :** Top qubit collapses to  $\alpha e^{i\pi/4}|0\rangle + \beta|1\rangle = e^{i\pi/4}T^\dagger|\psi\rangle$ . Apply Phase gate  $S$  to obtain  $e^{i\pi/4}T|\psi\rangle$ .

# Discussion Problems: CNOT from CZ and Hadamard

**Problem 1:** Show how to construct a Controlled-NOT (CNOT) gate using only CZ and Hadamard gates:

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This matches the action of a CNOT gate exactly!

## Problem 2: Toffoli Gate Basis Actions

- Flips target qubit if and only if both controls are  $|1\rangle$ :

$$\text{CCNOT } |110\rangle = |111\rangle, \quad \text{CCNOT } |111\rangle = |110\rangle$$

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## Problem 3: Deutsch Oracle for $f(x) = x$

- For  $f(x) = x$ , the oracle acts as a CNOT gate:

$$U_f |+\rangle |-\rangle = \text{CNOT } |+\rangle |-\rangle = |-\rangle |-\rangle$$

- This kicks back a phase of  $-1$  onto the control qubit, transforming it from  $|+\rangle$  to  $|-\rangle$ .

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- 3 Non-Clifford operations (like  $T$ ) are necessary for quantum advantage, and can be implemented fault-tolerantly using magic state distillation and injection.
- 4 Protocols like Bell state preparation, superdense coding, and Deutsch's algorithm highlight the power of quantum properties (entanglement, phase kickback) over classical systems.